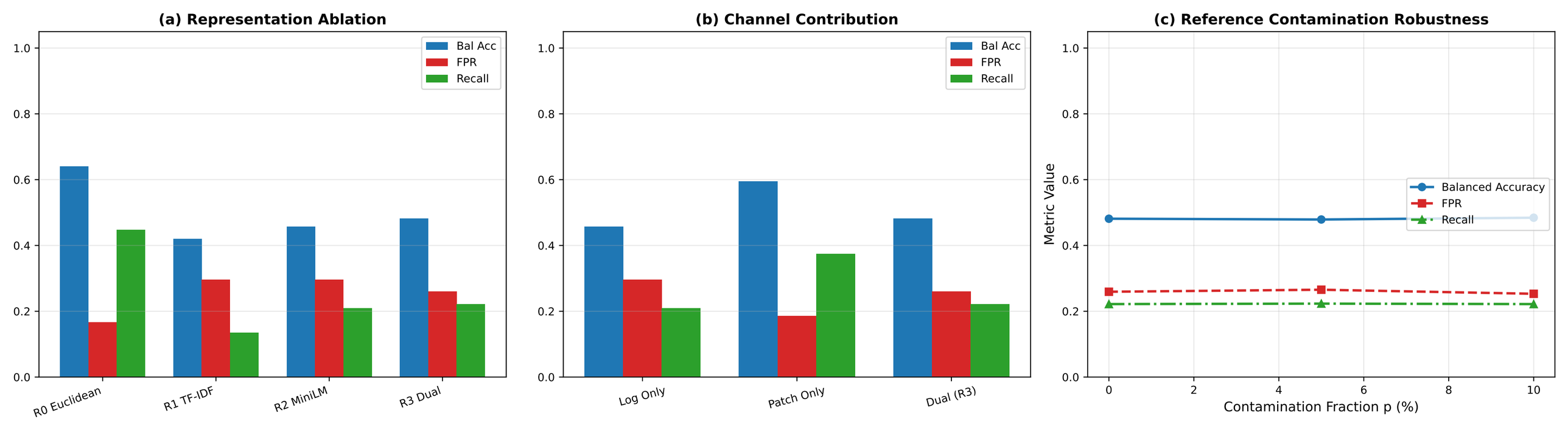
(a) Representation Ablation
Bal Acc
FPR
Recall
1.0
0.8
0.6
0.4
0.2
0.0
R0 Euclidean
R1 TF-IDF
R2 MiniLM
R3 Dual
(b) Channel Contribution
Bal Acc
FPR
Recall
1.0
0.8
0.6
0.4
0.2
0.0
Log Only
Patch Only
Dual (R3)
(c) Reference Contamination Robustness
Metric Value
Balanced Accuracy
FPR
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
Contamination Fraction p (%)